Exercise #7

Grid size = 19812mm x 16764mm
Unsuitable material = 450mm
Existing topsoil = 100mm
New topsoil = 150mm
Existing grades = 37.23; 38.01; 37.68
Finish grades = 38.50; 38.75; 38.00; 39.25
Cut or Fill-?

Area = 19.812m x 16.764m =
= 332.128m²
Avg. Existing grade = (37.23 + 38.01 + 37.68)/3 = 37.64 - (0.1 + 0.45) = 37.64 - 0.55 = 37.09m
Avg. Finish grade = (38.50 + 38.75 + 38.00 + 39.25)/4 = 38.625 - 0.15 = 38.475m
Avg. Existing grade = 37.09m < Avg. Finish grade = 38.475m
⇓
Fill = 38.475 - 37.09 =
= 1.385m
(depth)
Fill Req'd = Area x Depth =
= 332.128 x 1.385 = 460m³ x
x (compaction of 15%) =
= 460/0.85 = 541.2 = 542m³

Answer: Fill (type 3) = 542m³